Yuran Zhang

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Summary

BS degree in Computer Science & Statistics from McGill University with hands-on experience in software development, data analysis, and cloud infrastructure. Proficient in Python, Java, C++ and SQL, with a solid foundation and great passion in Machine Learning & NLP. Knowledgeable in scalable system design and experienced in developing and deploying AI models using ML frameworks to solve real-world challenges.

Canadian citizen- TN eligible

Education

McGill University *B.Sc. in Computer Science / Minor Statistics* *Montreal, QC, Canada* *Sep. 2021 – May 2025*

- ***Relevant courses:*** Data Structures & Algorithms, Software Design, Operating Systems, Probability & Statistical Analysis, Applied Regression, Machine Learning, NLP, Reinforcement Learning
- ***Leadership:*** Executive of the McGill Competitive Programming Club

Technical Skills

- **Programming Languages**: Python, Java, C++, R, SQL, Bash, JavaScript, HTML, CSS
- **Frameworks & Technologies**: Spring Boot, Microservice, Maven, REST APIs, Flask, Kafka, Redis, MySQL, PostgreSQL, MongoDB; AWS, MS Azure, Docker, Jenkins; ML libraries and tools (Numpy, Pandas, PyTorch, TensorFlow, Scikit-Learn)
- **Development Tools/Environment**: Git, GitHub, GitHub Actions, GitLab, IntelliJ IDEA, PyCharm, VS Code; Windows, Unix/Linux

Certifications

- **AWS Certified Cloud Practitioner**
- **ICPC Contest Certificate** (ranked **top 3 teams** at McGill, 18th place in ICPC Northeast NA Division 2024)

Work Experience

McGill University *Montreal, QC, Canada*

Lab Research Assistant (Volunteer) (Computational Biology & Healthcare / Machine Learning) *May 2025 – Present*

- Conducted **EDA** (Exploratory Data Analysis) and **visualization** on large scale EHR datasets (eICU, MIMIC-III) with **Jupyter Notebook**, and built **scalable preprocessing pipeline** (Python, Pandas, SQL) to clean, impute and standardize **200K+** patient records.
- Implemented and optimized **federated learning workflows** with LSTM/TCN models, reducing communication overhead by **4.5×** and improving **AUPRC** by **5%** across heterogeneous hospital datasets.

Eth Tech *Newark, CA, USA*

Software Engineer Intern *May 2024 - Aug. 2024*

Contributed to the **backend system development** of an **e-commerce platform** utilizing **microservices architecture** with **Java** based on **Spring Boot framework**.

- Engineered features for **enhanced product exploration, selection,** and **management**, utilizing **Spring Data** to streamline data operations across **MySQL** and **MongoDB databases**, resulting in a 30% improvement in query performance.
- Integrated a **Kafka system** for **real-time updates** on product availability, order status, and inventory management, supporting over 8,000 real-time events per day.
- Implemented a sophisticated **Redis Caching Mechanism** to optimize system performance, reducing response time by 60% and ensuring scalability to accommodate a 45% increase in peak user load.
- Developed an intelligent **recommender system** using a **deep learning-based Neural Collaborative Filtering approach** (NeuMF model) with **PyTorch** to analyze implicit user feedback and user-item interactions, delivering personalized product rankings and increasing click-through rates (CTR) by 25%.

YiSen Tech *Dalian, China*

Junior Software Developer Intern *May 2023 – Aug. 2023*

Contributed to the backend infrastructure for a **real-time chat web application** using Python with **Flask** and **WebSocket** protocols, supporting concurrent messaging for 300+ active users with sub-100ms latency.

- Built a containerized **microservice architecture** using **Docker**, implementing separate containers for message handling, user authentication, and notification services.
- Implemented asynchronous message processing with Python's **asyncio library** and integrated **Redis** for session management and message queuing, enabling real-time message delivery and reducing database load by 30%.
- Established **CI/CD pipeline** using **GitHub Actions** that automated container builds, ran unit tests with **Pytest**, and deployed containerized services to **AWS ECS**, while setting up **CloudWatch** monitoring for application performance tracking and error logging.

Selected Projects

Project for 2025 Canada HealthSync Innovation Challenge: AI-Driven Healthcare Decision Support System *Jan. 2025 – Mar. 2025*

Ranked ***top 5 team*** among 20+ teams

- Developed a hybrid ML model with **XGBoost** + **ClinicalBERT** to analyze structured lab data and unstructured clinical notes, achieving 85% precision in early risk prediction for cardiovascular events.
- Built a lightweight **RAG system on medical guidelines** (stored as text chunks in SQLite). Embedded queries with a distilled version of ClinicalBERT from HuggingFace, achieving high relevance (80%+) in retrieved documents.
- Containerized models with **Docker** and deployed as scalable **FastAPI microservices**; integrated FHIR-compliant APIs to sync with Supabase **PostgreSQL** for real-time data pipelines.